ABDULAZEEZ NURUDEEN ADEDOTUN

AI/ML Engineering & Backend Development | NLP & AI-Assisted Systems | 2+ Years Experience

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate with practical project experience in machine learning, natural language processing, AI-assisted software development, and backend engineering. Experienced in preparing datasets, developing and evaluating machine learning models, improving classification performance, and integrating technical solutions into user-facing applications.

Developed an NLP-based spam classification system using TF-IDF, Logistic Regression, and fine-tuned DistilBERT, achieving 99.10% accuracy and a 96.41% F1-score. Also developed a predictive maintenance model using the AI4I 2020 dataset, improving a Random Forest model's F1-score from 71.21% to 88.19% through feature engineering and model optimisation.

Additional experience includes contributing to AI-powered application development, backend integration, database-driven software, and an embedded machine-learning project involving sensor fusion, Random Forest classification, and a PyQt monitoring interface.

My technical interests include AI/ML engineering, AI evaluation, AI agent development, and backend engineering. I bring an engineering mindset, analytical problem-solving skills, collaborative development experience, and a strong commitment to continuous learning.

PROFESSIONAL EXPERIENCE

Nexus BuildLab — ML Intern / Backend Lead & Group Leader

July 2026 – October 2026 | Remote

- Contributing to the development of ScamCheck, an AI-powered application designed to identify potentially fraudulent messages
- Taking responsibility for backend development alongside machine learning internship responsibilities
- Working across the machine learning and application development workflow, collaborating with teammates to integrate technical components
- Coordinating team activities, tracking project progress, and supporting technical decision-making
- Collaborating with other team members on implementation, testing, and project delivery

Notely — AI / Backend Developer

December 2025 – June 2026 | Remote

- Contributing to the development of an AI-powered podcast note-taking application that supports timestamped notes and relevant episode information
- Working on transcript-processing and information-extraction workflows
- Contributing to AI-driven episode matching and confidence-scoring workflows
- Reviewing extracted information and generated outputs for relevance, accuracy, and quality
- Using Git and GitHub for version control, feature branches, pull requests, and collaborative code reviews

Transmission Company of Nigeria (TCN) — Industrial Trainee

Jul 2021 – Dec 2021; Jun 2024 – Sep 2024; Apr 2025 – Oct 2025 | Nigeria

- Gained practical exposure to electricity transmission operations, particularly Protection, Control and Metering
- Participated in preventive maintenance activities, equipment inspections, and operational monitoring
- Observed and supported relay testing and maintenance procedures

- Developed an understanding of transmission infrastructure, including 132kV and 33kV systems and 60MVA power transformers
- Worked in an environment that required adherence to technical procedures, workplace safety, and operational discipline

SELECTED PROJECTS

SMS Spam Detection — NLP & Transformer Classification

2026 | Nexus BuildLab

- Developed an SMS spam classification system using the SMS Spam Collection dataset
- Implemented a TF-IDF and Logistic Regression baseline and compared it with a fine-tuned DistilBERT transformer model
- Achieved 99.10% accuracy and a 96.41% F1-score with DistilBERT
- Reduced missed spam predictions from 30 with the baseline model to 4 with the transformer model in the reported evaluation
- Built a Streamlit interface and deployed the application using Hugging Face
- Considered model limitations, false predictions, privacy, and potential bias during evaluation

Predictive Maintenance — Machine Failure Prediction

2026 | Nexus BuildLab

- Developed a machine-learning classification pipeline using the AI4I 2020 Predictive Maintenance dataset, containing 10,000 records
- Conducted data preprocessing, exploratory analysis, feature engineering, and feature selection
- Addressed class imbalance and trained machine-learning models, including Random Forest
- Applied five-fold GridSearchCV and decision-threshold optimisation to improve model performance
- Improved the Random Forest F1-score from 71.21% to 88.19% and reduced false positives from 17 to 3 in the reported evaluation

ScamCheck — AI-Powered Scam Detection Application

2026 | Nexus BuildLab

- Contributing to a collaborative application designed to help users identify potentially fraudulent messages
- Working on the backend components and supporting integration with the machine-learning functionality
- Collaborating with the team to develop the application workflow and connect its technical components

Quantum Jersey — Full-Stack E-Commerce Platform

2024 – Ongoing | Personal Project

- Designed, developed, and launched an e-commerce platform for a sportswear retail business
- Implemented product catalogue and inventory-management functionality
- Built database-backed features using PostgreSQL and Supabase
- Configured authentication, Row Level Security (RLS), cloud storage, and product image management
- Integrated WhatsApp-based checkout to support customer ordering
- Deployed the application using Vercel

Auto Clinic — Automotive Service & Equipment Platform

2025 | Personal Project

- Built a web application for presenting automotive equipment and vehicle maintenance services
- Developed features for service and equipment information, maintenance-related workflows, and appointment booking

- Used AI-assisted development tools to support implementation and accelerate the development process

Underground IED Detection System Using Sensor Fusion

2025 – 2026 | B.Eng. Final-Year Project

- Designed and developed an underground explosive detection rover integrating magnetic, thermal, and gas sensors
- Processed sensor readings and applied sensor-fusion techniques to support identification of potentially hazardous objects beneath the ground
- Trained a Random Forest classification model and conducted a comparative evaluation against a shallow machine-learning model
- Developed a PyQt-based monitoring application to display live events and detection information transmitted from the rover during on-site operation
- Integrated the monitoring interface with the rover system to support real-time observation and assessment of detected ground objects
- Worked with Raspberry Pi and ESP32-based components as part of the embedded system architecture

EDUCATION

Bachelor of Engineering (B.Eng.) in Mechatronics Engineering

Air Force Institute of Technology (AFIT), Kaduna, Nigeria | 2022 – 2026

Classification: Second Class Upper

SKILLS & TOOLS

Machine Learning | Python | Software Development | Data Science | Backend Development | Natural Language Processing (NLP) | Scikit-learn | Transformers | DistilBERT | Random Forest | Logistic Regression | Pandas | NumPy | SQL | FastAPI | REST APIs | PostgreSQL | Supabase | Git | GitHub | Streamlit | Hugging Face | Linux | Vercel | AI Agents | Agentic Workflows | PyQt | Raspberry Pi | Sensor Fusion | MATLAB | SOLIDWORKS

CERTIFICATIONS

- Mercor — Introduction to Agents
- Google Data Analytics Professional Certificate

HOBBIES & INTERESTS

Artificial Intelligence, Machine Learning, Technology, Engineering, Football, Sports

LANGUAGES

English — Fluent

LINKS

LinkedIn: https://www.linkedin.com/in/nurudeen-abdulazeez

GitHub: https://github.com/Quantumdata66

Portfolio: https://quantumdata66.github.io/My-Portfolio